Formal Logic 1

Christopher R. Walker

September 28, 2026

1 Week 15: FOL Proofs

Proofs for FOL will be governed by the same rule system as TFL, with the addition of some new rules for dealing with singular terms, predicates and quantifiers. Of course, these rules will take the form of our familiar introduction and elimination rules, plus some derived rules that let us avoid repeating frequently needed subproofs.

Given the definitions for substitution instances discussed last week, defining elimination rules for $eforallx$ and introduction rules for $eexistsx$ is particularly straight-forward.[1].

Rule 1.1 (Universal Elimination (∀E)):

n	$\forall\gamma\ \phi\ (\dots\gamma\dots\gamma\dots)$	
	$\phi\ (\dots\alpha\dots\alpha\dots)$	$\forall$E, n

With existential introduction, we can even substitute the variable for any one or more instances of the relevant name, which we will depict as $\exists\gamma\phi(\dots\gamma\dots\alpha\dots)$.

Rule 1.2 (Existential Introduction (∃I)):

n	$\phi\ (\dots\alpha\dots\alpha\dots)$	
	$\exists\gamma\ \phi\ (\dots\gamma\dots\alpha\dots)$	$\exists$E, n

The two complementary operations (*existential elimination* and *universal introduction*) are more contrained in their definitions. For universal introduction (∀I), we should only allow the rule to apply when the name in question is *arbitrary*. For the purposes of proofs, this amounts to not being found in premises or unresolved subproofs since their use there makes them non-arbitrary.

Rule 1.3 (Universal Introduction (∀I)):

n	$\phi\ (\dots\alpha\dots\alpha\dots)$	
	$\forall\gamma\ \phi\ (\dots\gamma\dots\gamma\dots)$	$\exists$E, n

But only if α does not occur in a premise or an undischarged assumption.

Rule 1.4 (Existential Elimination (∃E)):

n	$\exists\gamma\ \phi\ (\dots\gamma\dots\gamma\dots)$	
i	$\phi\ (\dots\alpha\dots\alpha\dots)$	AS
j	ψ	
	ψ	$\exists$E, n, i–j

[1] In our rules, we let γ be metavariables for FOL variables and α be a metavariable for FOL names

But α must not occur in:

1. any assumption discharged before line i
2. $\exists\gamma\phi(\ldots\gamma\ldots\gamma\ldots)$
3. ψ

Similarly, proofs in FOL will proceed with a strategy based on the one we devised for FOL, with the addition of some new techniques to deal with quantifiers. Quite tersely, we can describe these new techniques as follows.

1. Work forward when dealing with the existential quantifier ($\exists$)
2. Work backward when dealing with universal quantifiers ($\forall$)

Like conjunction and disjunction, existential and universal quantifiers are mutually-definable with the help of the negation operator.

Rule 1.5 (Conversion of Quantifiers (CQ))): We can relate the quantifiers of first-order logic, with the following two pairs of complementary rules. One pair for handling a negated existential quantifier:

n			n		
	$\forall\gamma\ \neg\phi$			$\neg\exists\gamma\ \phi$	
	$\neg\exists\gamma\ \phi$	CQ, n		$\forall\gamma\ \neg\phi$	CQ, n

and one pair for handling a negated universal quantifier:

n			n		
	$\exists\gamma\ \neg\phi$			$\neg\forall\gamma\ \phi$	
	$\neg\forall\gamma\ \phi$	CQ, n		$\exists\gamma\ \neg\phi$	CQ, n

Finally, we will need some rules for handling the identity predicate (=). As usual, these will take to form of introduction and elimination rules for =.

Rule 1.6 (Identity Introduction (=I)):

	$\alpha =\alpha$	=I

And, as with existential introduction, we can perform *identity elimination* on any one or more instances of the relevant term.

Rule 1.7 (Leibniz's Law (=E)): Since identity is a symmetric relation, we define a pair of rules for the elimination of non-trivial identity sentences like $\alpha = \delta$.

m	$\alpha = \delta$		m	$\alpha = \delta$	
n	$\phi\,(\ldots\alpha\ldots\alpha\ldots)$		n	$\phi\,(\ldots\delta\ldots\delta\ldots)$	
	$\phi\,(\ldots\delta\ldots\alpha\ldots)$	=E, m, n		$\phi\,(\ldots\alpha\ldots\delta\ldots)$	=E, m, n

Without getting too deep into the metalogic, we can make some basic claims about first-order logic. The most important observation is that FOL is both *sound* and *complete*.

Definition 1.1 (Soundness and Completeness of First-Order Logic)**.** $ephi_1, \ldots, \phi_n \vdash \psi$ iff $ephi_1, \ldots, \phi_n \vDash \psi$

Corollary to this observation is the practical equivalence of several analogous proof-theoretic and model-theoretic concepts.

Definition 1.2 (Properties of First-Order Logic)**.** Since first-order logic is sound and complete, the following pairs of concepts are analogues.

model-theoretic	**iff**	**proof-theoretic**
valid	=	provable
validity	=	theorem
equivalent	=	interderivable
jointly satisfiable	=	jointly equivalent

Strategies for proving these concepts in FOL metalogic can be summarized in the usual table.

	yes	**no**
valid (sentence)?	prove $\vdash \phi$	find ϕ counter-model
contradiction?	prove $\phi \vdash \bot$	find $\neg\phi$ counter-model
equivalent?	prove $\phi \vdash \psi$ and $\psi \vdash \phi$	find model where ϕ and ψ differ
jointly satisfiable?	model where ϕ_i are all true	prove $\phi_1 \ldots \phi_n \vdash \bot$
valid (argument)?	prove $\phi_1 \ldots \phi_n \vdash \psi$	find model with $\phi_1 \ldots \phi_n$ all true but ψ false

References

Magnus, P., Button, T., Trueman, R., & Zach, R. (2025). *Forall x: Calgary: An introduction to formal logic*. University of Calgary. https://openlogicproject.org/